

Now *chroot* into that partition by running the command given below:

```
$sudo chroot /media/sda1
```

Then re-install GRUB, as follows:

```
#grub-install /dev/sda
```

The output should be like what's shown below:

```
Installation finished. No error reported.
```

If you get an error, then try the following command:

```
#grub-install --recheck /dev/sda
```

After a successful installation, exit from *chroot* and *umount* the file systems that were mounted to recover GRUB. Now reboot.

```
#exit
```

```
$sudo umount /media/sda1/proc
```

```
$sudo umount /media/sda1/dev
```

```
$sudo umount /media/sda1
```

```
$sudo reboot
```

You've successfully completed recovering your GRUB boot loader.

—Kousik Maiti,
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Find out the elapsed time of a running process

There are a lot of processes running on your Linux system. Here is a command that will let you know how long the process has been running:

```
#ps -eo "%p %c %t"|grep "sshd"
```

In response to the above command, you will get the following output:

```
2850 sshd      172-01:37:22
29532 sshd      125-09:07:10
```

In the above command %p is pid, %c is command and %t is elapsed time.

—Ravikumar R,
ravikumar.raam@gmail.com

Mounting a remote filesystem with ssh

Although running the *scp* or *ssh* command is very convenient, mounting the remote servers filesystem on the local system has solved many problems for me.

Here is how this can be done. To mount the remote filesystem to a local Linux workstation, you need to install the *sshfs*. To install, you can choose to use the package manager of your distro or you can compile and install manually. More details can be found at <http://fuse.sourceforge.net/sshfs.html>

After the installation is complete, you can run the following command on your Linux workstation:

```
sshfs username@hostname:/<Directory_path_on_Server>
/<mount_point>
```

That's it, so enjoy! (Of course, you should have the required permissions.)

—Sandesh Nagrare,
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Low-level HDD formatting

You often have a hard disk drive that needs to be formatted such that all data in it gets destroyed. Here is a command that can erase all data from your HDD.

```
#dd if=/dev/zero of=/dev/hda bs=1M
```

This will format your primary HDD (*hda*) to a *low level* using *dd*. This will fill the entire disk with a sequence of zeros. In case you want to mix zeros and ones to fill the HDD, you can use the following command:

```
#dd if=/dev/urandom of=/dev/hda bs=1M
```



Note: Please be very careful while using *dd* as the wrong use of this command can erase all your valuable data.

—Kousik Maiti,
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END

Share Your Linux Recipes!

The joy of using Linux is in finding ways to get around problems—take them head on, defeat them! We invite you to share your tips and tricks with us for publication in *OSFY* so that they can reach a wider audience. Your tips could be related to administration, programming, troubleshooting or general tweaking. Submit them at www.linuxforu.com. The sender of each published tip will get a T-shirt.